

A Method for Detecting Football Player Actions Using IMUs attached on Shin Guards

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ABSTRACT

In football matches, players perform various actions such as running, walking, passing, trapping, and shooting. Analyzing these actions is crucial for players to assess their performance and strategize effectively. Currently, the standard method for analyzing these actions involves setting up multiple cameras in the stadium to record the match and then analyzing the video footage obtained. However, this method is costly and is mainly accessible to professional teams with significant financial resources. It is used in specific environments such as training facilities and stadiums. There is a need for a system that can be easily used by amateur football players. This study proposes a method for analyzing the actions of football players using IMUs (Inertial Measurement Units) attached to both legs. The evaluation of the proposed method revealed that it was feasible to detect three actions - passes with the right foot, passes with the left foot, and shots with the dominant foot with an average F-score of 0.884 for three subjects.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

football; soccer; activity recognition; wearable sensors; accelerometer; gyroscope; IMU

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1 INTRODUCTION

In football matches, players perform various actions like running, walking, passing, trapping, and shooting. Analyzing these actions can help improve players' skills by allowing them to assess their performance. The frequency of passes and shots is a crucial factor in determining match outcomes in the UEFA Champions League[1].

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Understanding how often these actions occur during a game is beneficial for strategizing team tactics, whether at the professional or amateur level. Analyzing these actions typically involves installing multiple cameras in the stadium to capture footage, which is then analyzed through image processing. However, this method is costly due to the installation of high-performance video cameras, making it accessible only to professional teams with significant financial resources. Therefore, there is a need to develop a more affordable system for amateur football players. In this study, we propose a method for detecting actions such as passes and shots using inertial measurement units (IMUs) to capture the movements of football players.

2 SPORT SYSTEMS AND EXPERIENCES

2.1 good example

The first example is a wearable fitness tracker. Smartwatches, for instance, monitor activity levels during exercise and daily life, providing real-time feedback. Therefore, in sports where maintaining a specific heart rate zone is important, such as aerobic exercises like running and cycling, they can monitor heart rate in real time and give feedback through vibrations or displays when the heart rate zone changes. The advantages of this example are that feedback is provided in real-time and the convenience of simply wearing it on the wrist. The second example is a baseball training system developed by Mizuno Corporation. This system, called MA-Q, uses sensors embedded in the ball to measure spin rate, velocity, and spin axis. Additionally, with the dedicated application, results can be obtained in real time, providing feedback for each pitch. This allows players' pitching techniques, which were previously understood only through intuition, to be visualized. By scientifically analyzing their form, the quality of training can be improved.

2.2 bad example

The first example is a training system using VR. VR training allows for repetitive practice regardless of time and place and can simulate various situations to train decision-making skills. However, while VR provides visual feedback, it is difficult to replicate the exact feeling of the actual competition environment completely. Specifically, fine movements and force adjustments may differ from real-life feedback. Additionally, the implementation of VR systems has drawbacks such as cost and the complexity of using the system. Moreover, some people may experience VR sickness, which can lead to physical discomfort. It is considered effective to use a hybrid training approach where simulations are performed with VR and then the same movements are practiced in the actual competition environment, allowing for the acquisition of skills while experiencing the differences between the two.

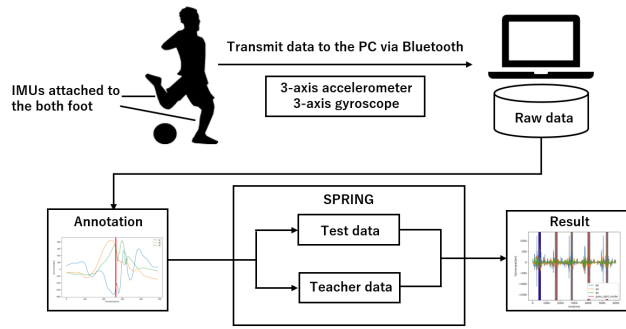


Figure 1: The overview of the proposed method.

The second example is a system that uses multiple cameras installed in a stadium to capture footage, which is then analyzed to obtain the positions of the ball, players, and their postures. This system can datafy the plays occurring on the field, making it useful for managing each player’s condition and devising team strategies. However, this method incurs enormous costs due to the installation of numerous video cameras, limiting its use to professional teams with ample financial resources. Therefore, there is a need to develop a system that amateur soccer players and enthusiasts can easily use on an individual basis to improve their skills. This research proposes a method to capture soccer players’ movements using IMUs and detect actions such as passes and shots from the obtained data.

3 PROPOSED METHOD

This section describes a method for detecting specific actions based on time-series data obtained from IMUs attached to both legs of football players.

3.1 System configuration

The diagram in Figure 1 shows the flow of the proposed method. The method involves three main steps: data acquisition using IMUs, preprocessing by annotating collected data, and detecting specific actions using SPRING[2], an algorithm that applies DTW for stream data. IMUs attached to both legs of the subject gather acceleration and angular velocity in the 3-axis, and this data is sent to a PC via Bluetooth. Afterward, the data is annotated based on simultaneously recorded video data to identify actions. The annotated data is then divided into training and test data. SPRING is applied to the test data to find waveforms similar to the training data, and the detection results are output. Detailed explanations of each process will be provided in the subsequent sections.

3.2 Data collection experiment

In this experiment, we utilized the AMWS020 IMU, which is manufactured by ATR-Promotions, for data acquisition. The AMWS020 features a 3-axis accelerometer, gyroscope, and magnetometer, and it can transmit data to a PC via Bluetooth. For this experiment, we configured the accelerometer range to $\pm 30G$, the gyroscope range to $\pm 4000dps$, and the sampling frequency to 200 Hz. The AMWS020 was securely attached to both shins. This positioning

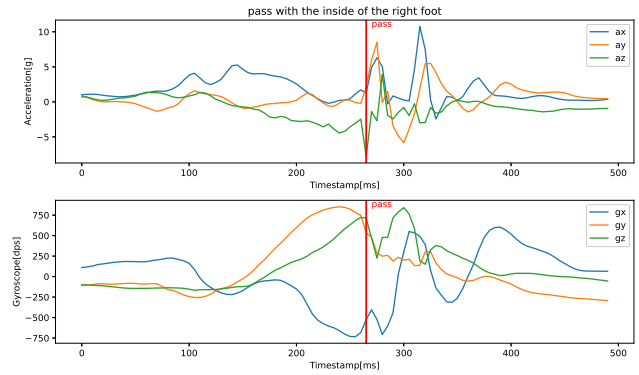


Figure 2: Waveforms of acceleration and angular velocity during passing.

aligns with the shin guard’s placement and does not hinder the player’s movements. During the experiment, participants executed face-to-face left and right inside passes and shots with a stationary ball, each performed 15 times and served as part of the training data collection. Furthermore, practical actions were executed for the test data collection. Specifically, participant A passed the ball to participant B, then ran diagonally forward while participant B passed the ball back. Participant A either passed the ball directly to participant C or took a shot while on the move. This sequence of actions was repeated 20 times for left and right inside passes and dominant foot shots, and the collected data was designated as test data. The experiment was recorded with a smartphone at 60 fps with 1080p resolution for annotation.

3.3 Annotaion

In this study, annotations were done using SyncPlay, a software manufactured by ATR-Promotions. This software allows simultaneous playback of time-series data and video data, making it possible to add arbitrary labels. To synchronize the time-series data obtained from the IMU and the video data, the subject intentionally created a peak by jumping at the start of the session. Then, using SyncPlay, the time when the peak in the time-series data was detected was aligned with the time of the jump to synchronize the data. A single annotator manually performed the annotation work. The results of annotating a pass with the right foot are shown in Figure 2. In the figure, the red line indicates the moment of contact with the ball. At the moment of ball contact, characteristic patterns appear in the acceleration and angular velocity waveforms, suggesting that detecting these waveforms from the obtained time-series data can identify actions such as passes.

3.4 Action detection

In this study, SPRING [2], which applies Dynamic Time Warping (DTW) for streaming data, was used to detect specific actions from the obtained time-series data. SPRING is an algorithm that comprehensively detects intervals (subsequences) similar to pre-provided training data in time-series data input from sources such as accelerometers and temperature sensors.

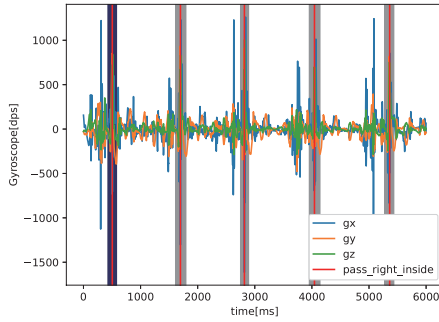


Figure 3: Example of action detection using SPRING.

SPRING calculates the distance between the input time-series data and the training data, and if the distance d_m satisfies the condition $d_m \leq \epsilon$ and also satisfies the condition $d_m \leq d_{min}$, it updates the value of d_{min} to d_m . At this time, by simultaneously updating the start point of the subsequence to $t_s = s_m$ and the endpoint to $t_e = t$, the interval $xt_s, \dots, xt_e$ is detected as the subsequence. Here, let ϵ be an arbitrary positive real number serving as the distance threshold. In this study, different thresholds were set depending on the subjects and types of actions. Since there are overlapping detected subsequences, the subsequence with the smallest distance to the training data among the overlapping intervals is output as a result. Specifically, by outputting the subsequence that satisfies the condition $\forall i; (d_i \geq d_{min} \vee s_i < t_e)$, the subsequence with the smallest distance to the training data among the overlapping intervals is output.

SPRING was applied to the 3-axis angular velocities $g_x(t), g_y(t), g_z(t) (t = 0, \dots, T)$ obtained from the data collection experiment. Among the detected subsequences, those detected on multiple axes were output as the recognition result. An example of detecting the inside pass with the right foot using SPRING is shown in Figure 3. The range indicated in navy is the training data, and the subsequences detected by SPRING are similar to the training data shown in gray. The red line represents the annotated correct label, and in this example, the subsequence detected by SPRING matches the correct label, correctly detecting inside passes with the right foot. In this study, specific actions are detected using SPRING in this manner, and the accuracy is evaluated.

4 EVALUATION

This section describes the evaluation experiment conducted to verify the effectiveness of the proposed method.

4.1 Threshold determination

This section describes the preliminary experiment to determine the threshold ϵ mentioned in 3.4. Three subjects participated in this experiment, and all subjects were right-footed. From the 15 training data samples described in 3.2, the first instance of each action was extracted as the training data for threshold determination. Then, by applying SPRING to the remaining data while changing the threshold by increments of 100,000, the threshold that resulted

in the highest F-value was determined as the optimal threshold. This process was conducted for each action performed by the three subjects, A through C, to determine the optimal threshold for each subject's action.

4.2 Evaluation experiment

In the evaluation experiment, SPRING was applied to the training data and test data to verify the extent to which subsequences similar to the training data could be detected from the test data. The first instance of each action extracted in the preliminary experiment for threshold determination was used as the training data in this experiment. Precision, recall, and F-score were calculated as evaluation metrics.

4.3 Results and discussion

The Tables 1, 2, and 3 show the results of applying SPRING to the test data of Subjects A, B, and C, respectively, detecting three types of actions: inside pass with the right foot, inside pass with the left foot, and shot with the dominant foot. The inside pass with the right foot was detected with an average F-score of 0.844, the inside pass with the left foot with an average F-score of 0.856, and the shot with the dominant foot with an average F-score of 0.952. The overall average F-score was 0.884. The inside pass with the right foot was performed as the initial pass during shot data collection, resulting in more detections than other actions. For Subject A, the F-score for right foot passes was the lowest among all subjects. In Subject A's data, there were seven instances where shots were incorrectly detected as passes, which likely reduced the precision and consequently lowered the F-score. Additionally, the F-score for Subject B's left foot passes was the lowest among all subjects. Since the left foot is Subject B's non-dominant foot, many passes were made from unstable positions, as observed in the footage. This led to several actions going undetected, resulting in a lower recall and consequently a lower F-score. Moreover, this time, the first instance of each action was used as the training data for the evaluation experiment. However, using different data as the training data could potentially improve accuracy.

5 CONCLUSION

We proposed a method to detect passes with the left and right foot and shots with the dominant foot, using angular velocity data gathered from IMUs attached to both legs of football players. The method collected data using AMWS020 for passes with both the left and right feet and shots with the dominant foot. The collected data was then divided into training and test datasets. The initial experiment established thresholds for applying SPRING using the training data. In the evaluation experiment, SPRING was applied to the test data using one sample from the training data as the ground truth. The specific actions were detected from the test data, resulting in an average F-score of 0.844 for passes with the right foot, 0.856 for passes with the left foot, and 0.952 for shots with the dominant foot.

REFERENCES

- [1] Carlos Lago-Peñas, Joaquín Lago-Ballesteros, and Ezequiel Rey. 2011. Differences in performance indicators between winning and losing teams in the UEFA Champions League. *Journal of human kinetics* 27, 2011 (2011), 135–146.

Table 1: Results of action detection for subject A.

Actions	True segments	Detected segments	True positives	False positives	False negatives	Precision	Recall	F-score
Pass with the right foot	63	68	53	15	10	0.779	0.841	0.809
Pass with the left foot	40	39	37	2	3	0.949	0.925	0.938
Shot with the dominant foot	20	20	20	0	0	1.00	1.00	1.00

Table 2: Results of action detection for subject B.

Actions	True segments	Detected segments	True positives	False positives	False negatives	Precision	Recall	F-score
Pass with the right foot	60	64	54	10	6	0.846	0.900	0.871
Pass with the left foot	40	36	28	8	12	0.778	0.700	0.737
Shot with the dominant foot	21	21	19	2	2	0.905	0.905	0.905

Table 3: Results of action detection for subject C.

Actions	True segments	Detected segments	True positives	False positives	False negatives	Precision	Recall	F-score
Pass with the right foot	62	72	57	15	5	0.792	0.919	0.851
Pass with the left foot	40	45	38	7	2	0.844	0.950	0.894
Shot with the dominant foot	21	19	19	0	2	1.000	0.905	0.950

[2] Yasushi Sakurai, Christos Faloutsos, and Masashi Yamamuro. 2006. Stream monitoring under the time warping distance. In *2007 IEEE 23rd International Conference*

on Data Engineering. IEEE, 1046–1055.